

Identification of Undocumented Meat Samples Seized at Heathrow Airport

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Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Background

Results

Conclusions

Introduction

Rarity status attributed to a species often increases its price and demand, with critically endangered species fetching the highest prices [1, 2]. The fallout of this is a positive feedback loop between rarity and demand which may drive an endangered species towards extinction and threaten whole ecosystems [3]. Species-level identification of undocumented meat samples is key in public health for validating the meat being delivered to consumers, but may also uncover networks carrying out illegal imports of endangered species, thus aiding conservation efforts by providing routes to intervene in the trade of these species.

If the morphology of a sample is compromised, DNA barcoding - comparing genetic sequences in a sample against known references - can aid in resolving the species of the sample [4]. Mitochondrial DNA (mtDNA) genes are commonly used for DNA barcoding because they occur in high copy numbers relative to nuclear DNA and typically lack recombination, keeping within-species diversity low which enables species-level identification [5]. Cytochrome c oxidase I (COI), found within mtDNA, is the primary barcode marker in the animal kingdom, and reference sequences for COI are abundant in databases such as BOLD and NCBI, permitting

barcoding against a diverse set of organisms [6, 7].

This analysis aims to resolve the species of four meat samples intercepted by UK Border Force officers at Heathrow Airport. The sequenced samples provide the starting point of this analysis pipeline.

Methods

Pipeline Overview

This pipeline consists of thirteen scripts which are used to take the provided sequence data (samples.zip) through to the visualisation of a phylogenetic tree for sample identification. The minimal input design of the pipeline and provided environment.yml file maximise reproducibility and minimise incorporating bias or human error into the analysis. The four main stages are described here.

QC and Pre-Processing

FASTQ sequence files were extracted from samples.zip and

BLAST

```
sort -k12,12g -k13,13gr -k5,5gr 5_blast_filtering/"$part_name"_blast.tsv \  
| awk -F'\t' '!seen[$3]++' > top_hits.tsv
```

Translation and Alignment

Tree Building and Viewing

Results

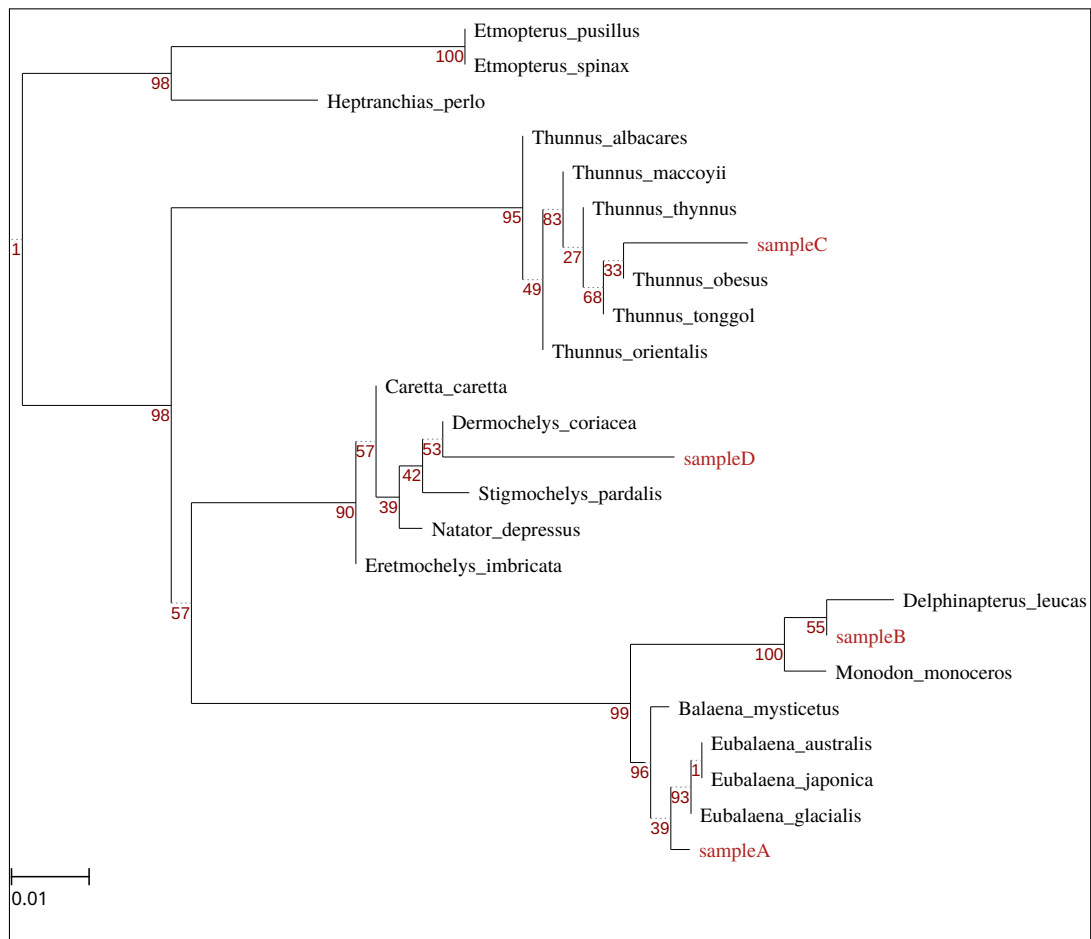


Figure 1: Maximum-likelihood phylogenetic tree with placement of unknown meat samples and top BLAST hits. Created with ete3 [8].

Discussion

Where cyt b and D-Loop sequences do exist, very few of these have been generated as part of forensic studies [23], with the vast majority submitted to databases as part of phylogenetic, molecular ecological and conservation studies [24], [25], [26]. Such sequence data can be generated and accepted into databases without any need for standard protocols or quality control, raising doubts over their suitability for forensic application [5]

However, unlike BOLD, GenBank does not store sequence chromatograms, collection metadata or photographs [6]. BOLD is a curation tool that also stores sequences, while GenBank is just a sequence repository. [9]

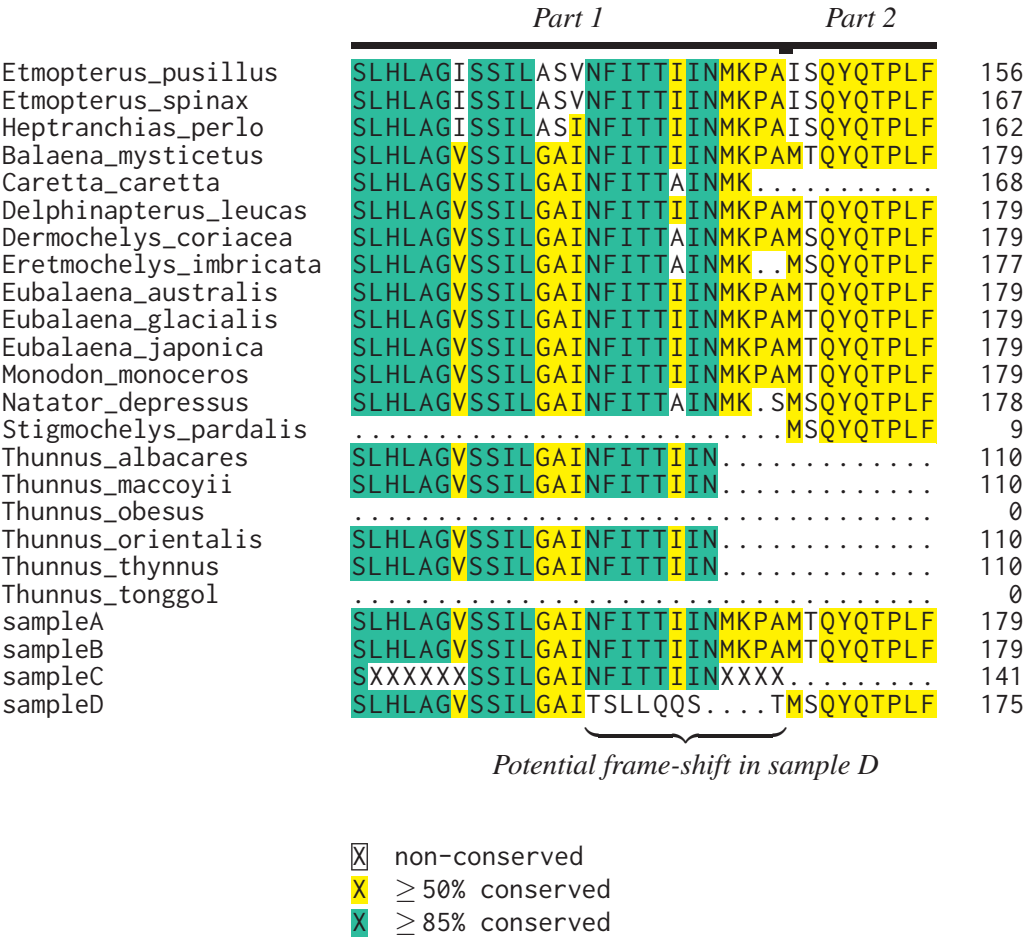


Figure 2: Window of supermatrix alignment with potential frame-shift error in sampleD near the highly-conserved NFITTIIN region.



Figure 3: Needleman-Wunsch global pairwise alignment at the tail end of sampleD_part1 against closest reference sequence. Alignment gap indicates a possible base miss in sampleD_part1 during sequencing.

References

[1] Yik-Hei Sung and Jonathan J Fong. Assessing consumer trends and illegal activity by monitoring the online wildlife trade. *Biological conservation*, 227:219–225, 2018.

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- [3] Liam J Hughes, Oscar Morton, Brett R Scheffers, and David P Edwards. The ecological drivers and consequences of wildlife trade. *Biological Reviews*, 98(3):775–791, 2023.
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- [5] Nick Dawnay, Rob Ogden, Ross McEwing, Gary R Carvalho, and Roger S Thorpe. Validation of the barcoding gene coi for use in forensic genetic species identification. *Forensic science international*, 173(1):1–6, 2007.
- [6] Teresita M Porter and Mehrdad Hajibabaei. Over 2.5 million coi sequences in genbank and growing. *PloS one*, 13(9):e0200177, 2018.
- [7] Sujeewan Ratnasingham, Catherine Wei, Dean Chan, Jireh Agda, Josh Agda, Liliana Ballesteros-Mejia, Hamza Ait Boutou, Zak Mohammad El Bastami, Eddie Ma, Ramya Manjunath, et al. Bold v4: A centralized bioinformatics platform for dna-based biodiversity data. In *DNA barcoding: Methods and protocols*, pages 403–441. Springer, 2024.
- [8] Jaime Huerta-Cepas, François Serra, and Peer Bork. Ete 3: reconstruction, analysis, and visualization of phylogenomic data. *Molecular biology and evolution*, 33(6):1635–1638, 2016.
- [9] Kelly A Meiklejohn, Natalie Damaso, and James M Robertson. Assessment of bold and genbank—their accuracy and reliability for the identification of biological materials. *PloS one*, 14(6):e0217084, 2019.

Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project:

https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures in Suspected Sequencing Error

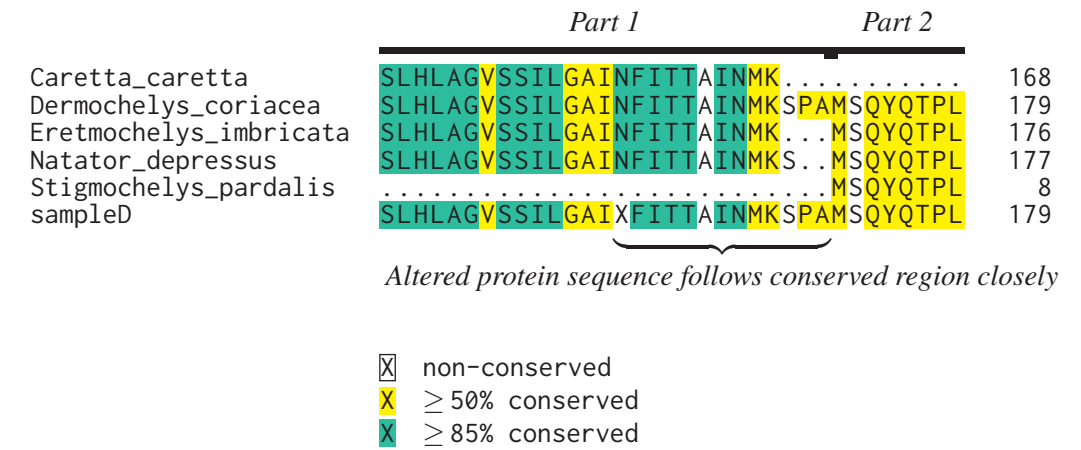


Figure 4: Subset of supermatrix alignment with sampleD and inner clade after inserting one ambiguous base (N) to the nucleotide sequence of sampleD.part1. This base resolves the apparent frame-shift, resulting in a protein sequence which aligns much more closely to highly-conserved NFITTIIN region compared to in Figure 2.

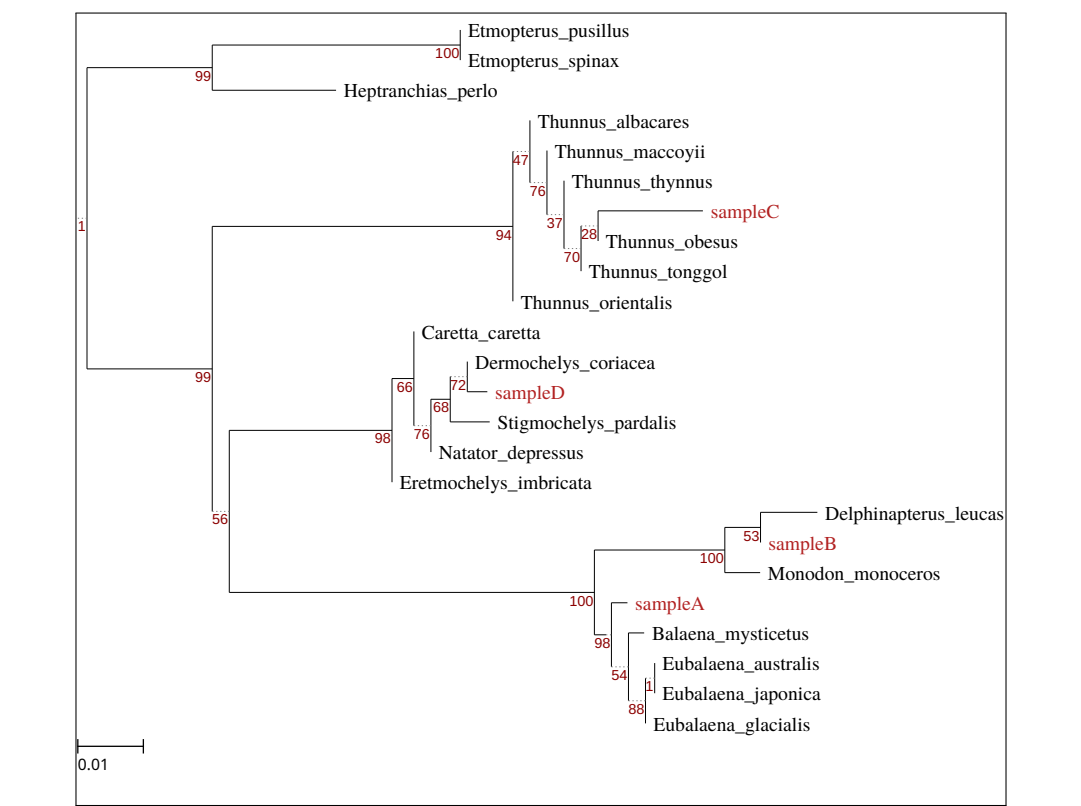


Figure 5: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into sampleD.part1. This tree is shown to demonstrate the impact of sequencing error and downstream frame-shift on branch lengths compared to Figure 1. Created with ete3 [8].